

Economic Analysis of Text and other Non-Standard Data

ISEG Summer School 2026

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6. Word Embeddings Without Neural Nets

Outline

Words Embedding with Local Contexts

Properties of Word Embeddings

How do I use Word Embeddings?

Bias in Language

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Word Embeddings

- ▶ “Word embeddings” often refer to GloVe – these are particular (popular) models for producing word embeddings.
 - ▶ **Previously:** focus on global document counts to predict an outcome
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 - rather than predicting some metadata, they predict the co-occurrence of neighboring words.
 - ▶ “You shall know a word by the company it keeps”
- ▶ From high-dimensional sparse representations to low-dimensional dense representations

Words and Contexts

A long line of NLP research aims to capture the distributional properties of words using a **word-context matrix M** :

- ▶ each row w represents a **word** of the vocabulary (e.g. “income”), each column c represents another word (e.g. “tax”) or linguistic **context** in which words can occur (e.g. “... pay corporate ___ tax to the relevant ...”)
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 - ▶ A matrix entry $M_{[w,c]}$ quantifies the strength of association between a word and a context in a large corpus.
 - ▶ Popular embeddings (such as glove) generally use 5- or 10-word context windows
- ▶ each word (row) $M_{[w,:]}$ gives a distribution over contexts.
 - ▶ different definitions of contexts and different measures of association → different types of **word vectors**.
 - ▶ these vectors have a **spatial interpretation** → geometric distances between word vectors reflect semantic distances between words.

GloVe Embeddings

- ▶ Pennington et al (2014) GloVe (Global Vectors for Word Representation) is embedding technique to represent words as dense vectors to capture semantic relationships between words
- ▶ Input: C_{ij} = local co-occurrence counts between words $i, j \in \{1, \dots, n_w\}$ within some co-occurrence window, e.g. ten words.

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Learn word vectors $\mathbf{w} = (w_1, \dots, w_i, \dots, w_{n_w})$, where $w_i \in (-1, 1)^{n_E}$, to solve

$$\min_{\mathbf{w}} \sum_{i,j} f(C_{ij}) \left(w_i^T w_j - \log(C_{ij}) \right)^2$$

where $f(\cdot)$ is weighting function to down-weight frequent words.

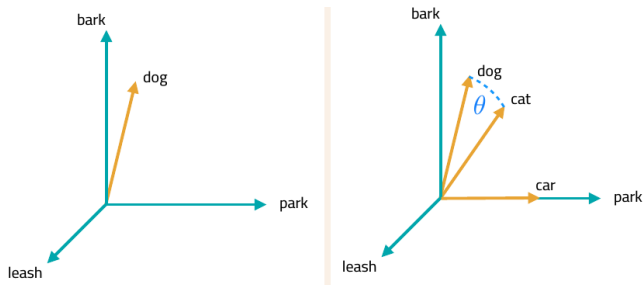
- ▶ Minimizes **squared difference** between:
 - ▶ **dot product of word vectors**, $w_i^T w_j$
 - ▶ **empirical co-occurrence**, $\log(C_{ij})$
[Arora et al (2016) put the PMI here instead of co-occurrence counts]
- ▶ Intuitively: words that co-occur should have high correlation (dot product)

Word2vec: From Counting to Predicting

- ▶ Like GloVe, word2vec starts from a simple idea: words appearing in similar contexts should have similar vectors.
- ▶ The main difference is how the vectors are learned:
 - ▶ **GloVe** uses a table counting how often words occur together.
 - ▶ **Word2vec** learns through a word-prediction task.
- ▶ Word2vec can use either of two prediction tasks:
 - ▶ **CBOW**: predict a word from the words around it.
 - ▶ **Skip-gram**: predict surrounding words from a given word.
- ▶ The model improves its word vectors whenever its predictions are wrong.
- ▶ After seeing many examples, words used in similar ways—such as *teacher* and *professor*—have similar vectors.

Word Similarity

- ▶ Once words are represented as vectors $\{v_1 = \mathbf{M}_{[w_1,:]}, v_2 = \mathbf{M}_{[w_2,:]}, \dots\}$, we can use linear algebra to understand the relationships between words:
 - ▶ Words that are geometrically close to each other are similar: e.g. “dog” and “cat”:



- ▶ The standard metric for comparing vectors is cosine similarity:

$$\cos \theta = \frac{v_1 \cdot v_2}{||v_1|| ||v_2||}$$

- ▶ Thanks to linearity, can compute similarities between groups of words by averaging the groups.

Word Vectors can produce Document Vectors

$$\vec{D} = \sum_{w \in D} a_w \vec{w}$$

- ▶ The “continuous bag of words” representation for document D is the sum, or the average (potentially weighted by a_w), of the vectors $\vec{w}$ for each word w in the document.
 - ▶ word vectors $\vec{w}$ constructed using GloVe (pre-trained or trained on the corpus).
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 - ▶ “Document” could be sentence, paragraph, section, etc.
- ▶ Arora, Liang, and Ma (2017) provide a “tough to beat baseline”, the SIF-weighted (“smoothed inverse frequency”) average of the vectors:

$$a_w = \frac{\alpha}{\alpha + p_w}$$

where p_w is the probability (frequency) of the word and $\alpha = .001$ is a smoothing parameter.

What do Word Embeddings Capture? (Budansky and Hirst, 2006)

- ▶ Semantic **similarity**: words sharing salient attributes / features
 - ▶ synonymy (car / automobile)
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- ▶ Word embeddings will recover one or both of these relations, depending on how contexts and vectors are constructed.

Most similar words to dog, depending on context window size

	2-word window	30-word window	
More paradigmatic		<u>kennel</u>	More syntagmatic
	cat	puppy	
	horse	pet	
	fox	bitch	
	pet	terrier	
	rabbit	rottweiler	
	pig	canine	
	animal	cat	
	mongrel	<u>bark</u>	
	sheep	alsatian	
	pigeon		

- ▶ Small windows pick up substitutable words; large windows pick up topics.

Parts of Speech and Phrases

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- ▶ Can improve the quality of embeddings in these cases by attaching the POS to the word (e.g. “like:verb”, “like:prep”) before training.
- ▶ The default model only works by word, but “new york $\neq$ ”new” + “york”
 - ▶ can tokenize phrases together (see Week 2 lecture) before training.

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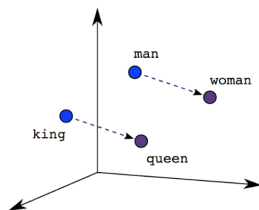
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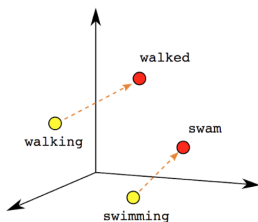
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 - ▶ And I don't see a solution to it.
- ▶ Relatedly, antonyms are often rated similarly, have to be careful with that.

Vector Directions $\leftrightarrow$ Meaning

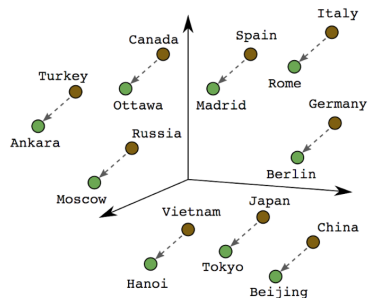
- ▶ Intriguingly, glove algebra can depict conceptual, analogical relationships between words:



Male-Female



Verb Tense



Country-Capital

Word Embeddings for Analogies

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- More generally: The analogy $a_1 : b_1 :: a_2 : b_2$ can be solved (that is, find b_2 given a_1, b_1, a_2) by

$$\arg \max_{b_2 \in V} \cos(b_2, a_2 - a_1 + b_1)$$

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- ▶ Often works better with normalized vectors (so that one long vector doesn't wash out the others)

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Tokenizing for Word Embeddings

- ▶ Need to think about what's the right way to pre-process data
 - ▶ drop capitalization
 - ▶ punctuation is optional
 - ▶ add special tokens for start of sentence and end of sentence
 - ▶ don't drop stopwords/function-words
 - ▶ for out-of-vocab words, substitute a special token or replace with part-of-speech tag

Which word-embedding?

- ▶ In many settings (e.g. a small corpus), better to use pre-trained embeddings.
- ▶ e.g, spaCy's GloVe embeddings:
 - ▶ one million vocabulary entries
 - ▶ 300-dimensional vectors
 - ▶ trained on the Common Crawl corpus
- ▶ Can initialize models with pre-trained embeddings, can fine-tune as needed.

“Enriching word vectors with subword information” (Bojanowski et al 2017)

- ▶ each word is represented as a bag of character n-grams. (e.g., spicy = (spi, pic, icy)).
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- ▶ learn embeddings for the character segments, and construct word embedding by summing over the segment embeddings
- ▶ competitive with glove in standard tasks; better in some languages.

Limitations

Standard word embeddings (e.g. glove) have a number of limitations:

- ▶ **n-grams**: does not produce embeddings for multi-word phrases
- ▶ **rare words**: a word that shows up just once or twice won't be well-defined
- ▶ **polysemy**: you get one vector for multiple senses of a word (e.g. “**glass** of water” vs “window **glass**”)

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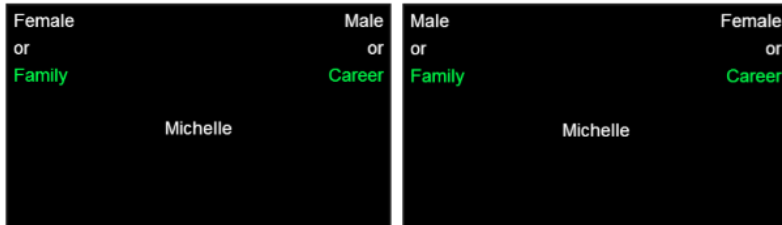
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- ▶ Subjects asked to assign words to categories (Greenwald et al. 1998)

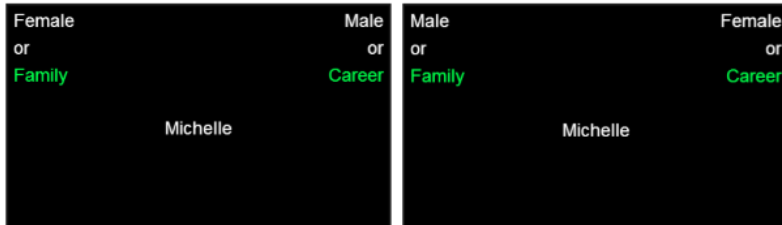


- ▶ Comparing reaction times across trials with different word pairs:
 - ▶ subjects tend to be slower and more error-prone in assignments against stereotype (e.g. "Michelle" goes to "Female or Career").

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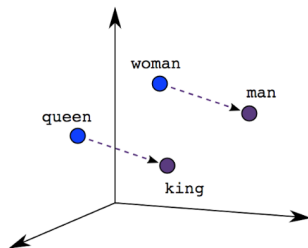
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 - ▶ IAT score = difference in reaction time between stereotype-consistent and stereotype-inconsistent rounds.

Caliskan, Bryson, and Narayanan (*Science* 2017)

- ▶ “We replicated a spectrum of known biases, as measured by the Implicit Association Test, using a widely used, purely statistical machine-learning model trained on a standard corpus of text from the World Wide Web. . . .”

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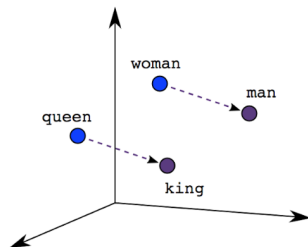


Analogies

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- ▶ walked : walking :: swam : swimming

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- ▶ **man : programmer :: woman : homemaker**
- ▶ **he : physician :: she : nurse**

Example Stimuli

- ▶ Targets:
 - ▶ **Flowers:** aster, clover, hyacinth, marigold, poppy, azalea, crocus, iris, orchid, rose, bluebell, daffodil, lilac, pansy, tulip, buttercup, daisy, lily, peony, violet, carnation, gladiola, magnolia, petunia, zinnia.
 - ▶ **Insects:** ant, caterpillar, flea, locust, spider, bedbug, centipede, fly, maggot, tarantula, bee, cockroach, gnat, mosquito, termite, beetle, cricket, hornet, moth, wasp, blackfly, dragonfly, horsefly, roach, weevil.

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- ▶ Attributes:
 - ▶ **Pleasant:** caress, freedom, health, love, peace, cheer, friend, heaven, loyal, pleasure, diamond, gentle, honest, lucky, rainbow, diploma, gift, honor, miracle, sunrise, family, happy, laughter, paradise, vacation.
 - ▶ **Unpleasant:** abuse, crash, filth, murder, sickness, accident, death, grief, poison, stink, assault, disaster, hatred, pollute, tragedy, divorce, jail, poverty, ugly, cancer, kill, rotten, vomit, agony, prison.

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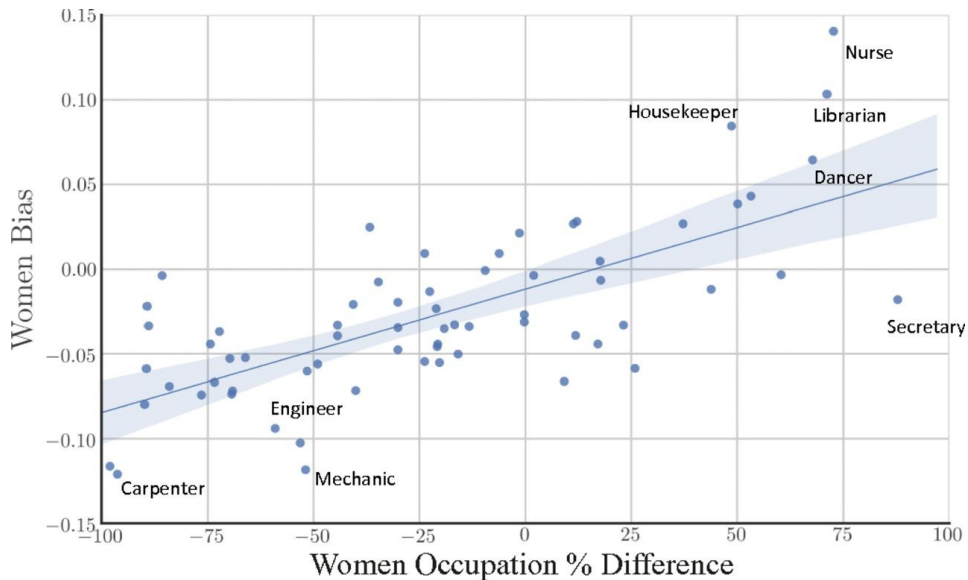
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 - ▶ European-American names vs. African-American names
- ▶ Male names vs. Female names:
 - ▶ Career words (e.g. professional, corporation, ...) vs. family words (e.g. home, children, ...)
 - ▶ Math/science words vs arts words

Garg, Schiebinger, Jurafsky, and Zou (PNAS 2018)



Women's occupation relative percentage vs. embedding bias in Google News vectors.